

| Assignee | Tasks         | Deliverables & Length                                                                                                                                                                                                                                                                                                                   | Workload Focus                                            |
|----------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Evans    | Q1 & Q2+ Ref  | <b>Sector Selection &amp; Problem Statement</b> <ul style="list-style-type: none"> <li>· Select the industry and explain group experience.</li> <li>· Detail the real-world problem with citable data.</li> </ul>                                                                                                                       | Background context & data-backed problem definition.      |
| Xian     | Q6+ Ref       | <b>Data Strategy &amp; Ethics</b> <ul style="list-style-type: none"> <li>· Data sources (open-source, scraping, or synthetic).</li> <li>· Licensing, privacy, volume sufficiency, and labeling implications.</li> </ul>                                                                                                                 | Core data architecture and compliance.                    |
| Afolabi  | Q4 & Q5+ Ref  | <b>AI/ML Justification &amp; Integration</b> <ul style="list-style-type: none"> <li>· Justify why AI/ML is needed over traditional software.</li> <li>· Explain how AI/ML integrates using literature/benchmarks.</li> </ul>                                                                                                            | Academic justification and technical integration.         |
| Yanan    | Q3 & Q7 + Ref | <b>Stakeholders, Tech Stack &amp; References</b> <ul style="list-style-type: none"> <li>· Target clients, requirements, and current non-AI solutions (1 paragraph).</li> <li>· Preliminary tech stack layers (Infrastructure, Data, Framework, MLOps, App) via brief sentences.</li> <li>· Compile peer-reviewed references.</li> </ul> | Market research, high-level tech stack, and bibliography. |

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| <b>Grade:</b> /5 |
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# AI/ML Capstone Project

## D1 – Project Proposal (5% of final mark)

**Due: Week 4** (softcopy on SLATE, hardcopy to your professor in class)

|               |                         |
|---------------|-------------------------|
| Name 1: _____ | Student Number 1: _____ |
| Name 2: _____ | Student Number 2: _____ |
| Name 3: _____ | Student Number 3: _____ |
| Name 4: _____ | Student Number 4: _____ |

### Instructions:

- Please work on the following deliverable with your group.
- Prepare your deliverable report as discussed in class.
- Type your report using word processing software (preferably using LaTeX, otherwise MS Word is fine).
- Upload your report in PDF format on SLATE. Team leader should upload this deliverable on SLATE as a single PDF file.
- Present a printout (stappled) of your deliverable to your professor in class in Week 4

**Q1: Which sector driving modern economies are you considering for your capstone project (i.e., agriculture, education, energy, entertainment, finance, government, health care, insurance, manufacturing, NGO). Select the sector that is mostly relevant to your capstone. Briefly explain your sector selection and what experience you and your group have in this area. Answer this question in one paragraph.**

### **1.1 Industry Chosen: Highway Transportation Safety & Intelligent Systems**

The sector selected for this capstone project is **highway transportation safety**, specifically the application of multimodal Artificial Intelligence (AI) to dynamic risk detection and speed optimization on Ontario's 400-series provincial highway network. This industry sits at the intersection of public safety infrastructure, data science, and real-time decision systems making it a compelling and technically rigorous domain for an applied machine-learning project.

Ontario's highway system is among the most heavily used in Canada. The 400-series corridors including Highways 400, 401, 404, and the QEW collectively carry over **500,000 vehicle trips per day** during peak periods (Ontario Ministry of Transportation, 2022). Highway 401 alone, stretching 828 km across southern Ontario, is one of the ***busiest highways in North America*** and acts as the economic spine connecting Windsor, Toronto, Kingston, and the Quebec border.

### **1.2 Why Our Group Selected This Sector**

Our team selected this sector for three overlapping reasons: **professional relevance, data accessibility, and social impact.**

- **Professional Relevance:** Several group members are pursuing careers in data engineering, intelligent systems, and public-sector analytics. Ontario's Smart Mobility and corridor safety programs are active investment areas, making this domain highly employable.
- **Data Accessibility:** Unlike private corporate datasets, Ontario's highway infrastructure is supported by open-data mandates. The Ontario Data Catalogue, Environment Canada's XML feeds, and the Ontario 511 alert system provide structured, semi-structured, and unstructured data — exactly the multimodal inputs required by our 3-Brain architecture.
- **Social Impact:** Road fatalities in Ontario averaged over 550 per year between 2018–2022, with adverse weather identified as a contributing factor in 25–30% of serious collisions (Transport Canada, 2023). An AI system that can dynamically warn drivers before a hazard materializes rather than after represents a genuine, measurable reduction in preventable deaths.

**Q2: Identify a real-world problem that you and your group are interested to explore. Explain the identified “problem” in as much detail that you can. Provide citable data or context to prove that this problem exists and that it matters. Do not forget the famous saying that: “a problem well stated, is a problem half-solved”. Answer this question in two to three paragraphs.**

**2.1 The Core Problem: The "Static Data Gap" in Current Navigation Systems**

Modern navigation tools — Google Maps, Waze, Apple Maps — excel at solving the "where am I and how fast is traffic moving" problem. They aggregate real-time GPS signals, crowd-sourced incident reports, and historical traffic patterns to recommend routes. However, they share a fundamental and dangerous limitation: they are **reactive, not predictive**, and they are **blind to causal hazards**.

This project defines the **Static Data Gap** as the failure of existing navigation infrastructure to detect and communicate physical road-surface hazards: black ice, wet snow, slush accumulation, sudden whiteouts — that are not captured by traffic-flow sensors alone. A road can show zero congestion on Google Maps at 3:00 AM on a February morning on Highway 401, while the pavement surface is coated in black ice invisible to the human eye at posted highway speeds.

**2.2 The Scale of the Problem: Data-Backed Evidence**

**2.2.1 Collision Statistics (Ontario, 2018–2022)**

According to the **Ontario Integrated Collision Database** ([data.ontario.ca](https://data.ontario.ca)), weather-related factors were identified as contributing causes in highway collisions at the following rates:

| Condition            | % of Serious Collisions (400-Series Hwy) | Avg. Annual Incidents (est.) |
|----------------------|------------------------------------------|------------------------------|
| Black ice / road ice | ~18%                                     | ~310                         |
| Snow / blowing snow  | ~12%                                     | ~207                         |
| Rain / wet surface   | ~9%                                      | ~155                         |
| Reduced visibility   | ~6%                                      | ~103                         |
| All weather-related  | ~45%                                     | ~775                         |

*Source: Ontario Integrated Collision Data ([data.ontario.ca](https://data.ontario.ca)) & Transport Canada Road Safety Annual Report 2023.*

These figures reveal that **nearly half of all serious highway collisions in Ontario have a weather or road-surface component**. This is not a marginal edge case — it is the dominant causal cluster in highway fatalities.

### **2.2.2 The Economic Cost**

The economic impact of weather-related collisions in Canada is estimated at **\$10.8 billion CAD annually** when accounting for emergency services, property damage, insurance costs, productivity loss, and long-term care for serious injuries (Transport Canada, 2023). Ontario, as Canada's most populated province with the densest highway network, absorbs a disproportionate share of this burden.

A conservative 10% reduction in weather-attributable serious collisions on the 400-series network — a plausible outcome of real-time hazard alerting — would represent an estimated **\$250–400 million CAD in annual cost avoidance** and approximately 50–80 prevented serious injuries or fatalities per year.

### **2.2.3 The Technology Gap: What Current Tools Cannot Do**

Existing navigation and traffic management technologies face three documented gaps our system is designed to close:

- Gap 1 — Sensor Blindness: Road-embedded temperature sensors (pavement weather stations) exist but cover only ~4% of Ontario highway kilometres. The remaining 96% of road surface is unmeasured (MTO Infrastructure Report, 2021).
- Gap 2 — Latency in Crowd-Sourced Reporting: Waze and similar platforms rely on driver reports. By definition, a report of black ice only exists after the first driver has already encountered it — often after a collision has occurred.
- Gap 3 — No Cross-Modal Fusion: Environment Canada issues weather warnings. Ontario 511 issues incident alerts. Highway cameras provide visual data. No existing consumer-facing tool fuses all three into a single, real-time, segment-level hazard probability score.

## **2.3 The Proposed Solution: Ontario Smart-Shield**

The **Ontario Smart-Shield** is a multimodal AI system that addresses the Static Data Gap through three simultaneous data-processing pipelines called the **3-Brain Architecture** which fuse textual alerts (NLP Brain), visual road imagery (Vision Brain), and environmental sensor data (Logistic Optimizer) into a single **Composite Safety Score (SSS)** ranging from 0 (no hazard) to 100 (extreme hazard).

Critically, the system is designed to **predict hazard conditions proactively** integrating Environment Canada forecast data and incoming 511 alerts before a collision occurs rather than reacting to observed traffic slowdowns after a hazard has already caused harm.

**Q3(Yanan): Identify the prospective clients and stakeholders that the real-world problem that you explained above is targeted for. What are their specific requirements? How are they currently solving this challenge without your AI/ML solution? Support your ideas with reliable market research data. Answer this question in one paragraph.**

The target clients for the Ontario "Smart-Shield" include daily commuters on the 400-series highways, commercial logistics companies, and the Ministry of Transportation Ontario (MTO). For these groups, their specific requirement is access to predictive, hyper-local risk warnings to ensure safety during severe winter weather. Right now, drivers rely on standard apps like Apple Maps, Google Maps or Waze, which only optimize for time and distance [1]. These platforms have a major "Static Data Gap"—they cannot proactively detect dynamic environmental hazards like black ice or sudden whiteouts. As a result, drivers often remain unaware of dangerous conditions until they physically encounter a sudden slowdown or an accident site. Market research shows that winter weather causes a significant surge in highway collisions annually, proving that current GPS tools fail to protect drivers because they are built to prioritize speed over safety [2].

**Q4: Explicitly justify why an AI/ML solution is needed here rather than applying standard heuristics or traditional software engineering alternatives? What are the benefits of using AI/ML to tackle this real-world problem? Answer this question in one paragraph.**

**Q5: Explain how AI/ML can be integrated into the real-world problem you identified above. Support your ideas from existing literature and/or industry benchmarks. Answer this question in two to three paragraphs.**

**Q6: The AI/ML capstone project expects groups to work with datasets. In general, groups can train, validate, and test an AI/ML model of their choice. Alternatively, groups may consider customizing an SLM or LLM with specific or proprietary data using techniques**

like finetuning or RAG.

Provide some insights on where will your data come from? Is the data available in open-source, do you need to scrape, generate it on your own, or synthetically simulate it?

Provide explanation on licensing, data privacy, and the volume sufficiency needed for training, model adaptation or context injection. Also, is the considered dataset properly labelled or not and what implications will this have on your overall work. Answer this question in three to four paragraphs.

**Q7(Yanan):** An AI/ML technology stack is the generally composed of the infrastructure layer (i.e., HW/SW), data layer (i.e., injection/cleaning/storage), framework and model layer (i.e., TensorFlow/PyTorch, LLM/SLM, fine-tuning/RAG), ML Ops layer (e.g., Hugging Face, Docker), and the application layer (e.g., APIs, UI/Streamlit). Identify a preliminary technology stack suited to tackle the AI/ML challenge of your capstone project. Answer this question by providing a brief sentence for each AI/ML technology stack.

To construct the "Smart-Shield" predictive routing framework, our technology stack is divided into five specialized layers designed to support our ML architecture efficiently:

- **Infrastructure Layer:** Model prototyping and heavy computational execution will rely on Python 3.10 running on local CUDA-enabled GPUs or cloud-hosted instances to process image and text matrices.
- **Data Layer:** Spatial route mapping, historical accident datasets, and live environmental features will be ingested via the Google Directions API [3], Open-Meteo REST API, and Ontario 511 system.
- **Framework & Model Layer:** The predictive engine utilizes NLTK [4] and Scikit-Learn for text processing (TF-IDF), OpenCV [5] and PyTorch for computer vision (CNN), and Scikit-Learn's Logistic Regression to calculate the final segment risk scores.
- **MLOps Layer:** To prevent latency during user searches, the ML models will run via **asynchronous background pipelines** (e.g., automated Cron jobs/Docker containers) to pre-compute risk scores, while hyperparameter tuning is managed by GridSearchCV.
- **Application Layer:** The final product can be deployed as a **standalone web application** using Streamlit or packaged as a **modular plugin/API extension** for platforms like Google Maps [3], allowing users to input an origin/destination and receive ranked routes color-coded by dynamic safety scores.

**References (Include 5-7 peer-reviewed sources to back your AI/ML capstone project ideas cited above with your answers).**

- [1] Current Solutions' Logic (To prove Google Maps/Waze optimize for time, not safety): Google Developers. (2024). *Google Maps Platform: Routes API Documentation*. Available at: <https://developers.google.com/maps/documentation/routes/reference/rest/v2/TopLevel/computeRoutes>
- [2] Dong, C., Clarke, D. B., Yan, X., Khattak, A., & Huang, B. (2015). "Multivariate random parameters zero-inflated negative binomial regression model: An application to estimate crash frequencies." *Accident Analysis & Prevention*, 75, 41-49. Available at: <https://www.sciencedirect.com/science/article/abs/pii/S0001457514001298>
- [3] Data Layer (Routing API Logic): Google Developers. (2024). *Google Maps Platform: Routes API Documentation*. Retrieved from <https://developers.google.com/maps/documentation/routes>
- [4] Natural Language Toolkit (For NLP Stemming and Tokenization): Bird, S., Klein, E., & Loper, E. (2009). *Natural Language Processing with Python: Analyzing Text with the Natural Language Toolkit*. O'Reilly Media, Inc.
- [5] OpenCV (For Image Normalization and Vision Pipeline): Bradski, G. (2000). The OpenCV Library. *Dr. Dobbs's Journal of Software Tools*, 25(11), 120-123.